

# Cable dimensioning

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## 1 NRS cable dimensions

Cables will have 10cm at each end to provide a service loop.

### 1.1 Voltage drop

Voltage drop across cables calculated from the following derived formula:

$$A = \frac{2 \times \rho(T) \times L \times I}{\Delta V}$$

Where  $L$  is the length of the cable in meters,  $\Delta V$  is the accepted voltage drop, and  $\rho(T)$  is:

$$\rho(T) = \rho_0 \times [1 + \alpha \times (T - T_0)]$$

Here  $\rho_0$  is the material resistivity,  $\alpha$  is the temperature coefficient for the material and  $T_0$  is the reference temperature for the material's resistivity.(I.E. the temperature where  $\rho$  of the material is equal to  $\rho_0$ )

```
[ ]: copper = {  
    'resistivity': 1.68e-8, # ohm-meters  
    'temperature_coefficient': 0.0039, # percentage per degree Celsius  
    'reference_temperature': 20 # degrees Celsius for the resistivity value  
}  
  
def calculate_wire_crossection(length: float,  
                               current: float,  
                               voltage_drop: float,  
                               temperature: float,  
                               material: dict = copper)-> float:  
  
    """  
    Calculate the required cross-sectional area of a wire.  
  
    Parameters:  
    - length (float): Length of the wire in meters.  
    - current (float): Current flowing through the wire in amperes.  
    - voltage_drop (float): Permissible voltage drop in volts.  
    - temperature (float): Operating temperature in degrees Celsius.  
    - material (dict): Material properties of the wire.  
    Returns:
```

```

- float: Required cross-sectional area in square millimeters.
"""

# Calculate resistivity at the given temperature
resistivity_at_temp = material['resistivity'] * (1 +
↳material['temperature_coefficient'] * (temperature -
↳material['reference_temperature']))

# Calculate the required cross-sectional area in square meters
area_in_square_meters = (2 * resistivity_at_temp * length * current) /
↳voltage_drop

# Convert the area to square millimeters (1 m^2 = 1e6 mm^2)
area_in_square_mm = area_in_square_meters * 1e6

return area_in_square_mm

```

## 1.2 HPRS

Servo

Setting the acceptable voltage drop  $\Delta V$  to  $0.2V$ ,  $L = 2.5M$ ,  $I_{Peak} = 2A$  and  $T = 60^\circ C$  gives us:

```

[ ]: length = 1.5 # in meters
length = length + 2*0.1 # add 10 cm on each side for connections and service
↳loops
current = 3 # in amperes
voltage_drop = 0.2 # in volts
temperature = 70 # in degrees Celsius
required_area = calculate_wire_crossection(length, current, voltage_drop,
↳temperature, material=copper)
print(f"Required cross-sectional area: {required_area:.2f} mm^2")

```

Required cross-sectional area:  $1.02 \text{ mm}^2$

## 1.3 MCRS

Setting the acceptable voltage drop  $\Delta V$  to  $0.2V$ ,  $L = 1M$ ,  $I_{Peak} = 2A$  and  $T = 80^\circ C$  gives us:

```

[ ]: length = 1 # in meters
length = length + 2*0.1 # add 10 cm on each side for connections and service
↳loops
current = 2 # in amperes
voltage_drop = 0.2 # in volts
temperature = 70 # in degrees Celsius
required_area = calculate_wire_crossection(length, current, voltage_drop,
↳temperature, material=copper)
print(f"Required cross-sectional area: {required_area:.2f} mm^2")

```

Required cross-sectional area:  $0.48 \text{ mm}^2$